

Mouse auditory experiment

A first plan

October 8, 2026

We want to play short tones to a mouse and record its brain responses. SONIC tested awake sheep; our mouse state and tone settings remain open. (Methods 6.6, p.13)

1. Choose the sounds

- Choose the frequency, duration, loudness, and gap between tones.
- **SONIC:** initially explored 1.4–32 kHz; benchmark tones were 10 ms sine waves with 2 ms fades at each end. (Methods 6.1–6.2, p.11)
- **SONIC:** random frequencies, balanced within each block, with no silence between tones. (Methods 6.1, p.11)
- Save the sequence before playback, as SONIC did. (Methods 6.2, p.11)

2. Set up the speaker

- Plan as though we need to assemble the whole sound setup.
- **SONIC:** NI USB-6361 output at 192 kHz, TDT SA1 amplifier, and MF1 free-field speaker; about 80 dB SPL at sheep-ear distance. (Methods 6.2, p.11)
- **Our plan:** compare the equipment on the next page, keep speaker placement fixed, and measure each tone with a calibrated microphone at the mouse-ear position.

3. Keep everything on one timeline

- Schedule each tone with a timing pulse and record it alongside the neural signals.
- **SONIC:** recorded electrical sound output and trial-start markers; shared clock references aligned these with Connexus brain recordings. (Methods 6.2, p.11)
- **Our additions:** synchronize a microphone recording to check actual sound arrival. Align video frame times too if we add behavior video.

Reference: Perkins et al. (2025), SONIC: A Benchmarking Paradigm for Brain-Computer Interfaces, bioRxiv v1. Page numbers above refer to the printed manuscript. Mouse choices and additions are proposals.

Equipment we need

Mouse setup: building from scratch

Xiaorong forwarded Tinghan's equipment list. We will use it to compare options; access to those items still needs confirming.

1. Play the tones

- **Candidate path:** computer $\rightarrow$ timed sound output (DAC/DAQ) $\rightarrow$ ED1 driver $\rightarrow$ ES1 speaker.
- **ES1:** plays sound through the air. Its intended range starts at 4 kHz; SONIC's lowest tones fall below that range. (TDT speaker specs)
- **ED1:** also needs zBus power, such as ZB1PS, and the matching speaker cable. It does not generate the tones. (TDT driver specs)
- **Other choices:** EC1 delivers sound through tubing. Peerless XT25TG30-04 and Scan-Speak R2004/602200 need a conventional amplifier; check their usable frequency range before choosing.

2. Measure the sound

- **Avisoft option:** the complete UltraSoundGate 116Hm kit includes a CM16/CMPA microphone and preamplifier. The 116Hm alone is the recording interface. (Avisoft kit details)
- **B&K option:** the 4939 microphone needs a compatible preamplifier, polarization supply, and recorder. (4939 datasheet)
- Either path needs calibration records and a sound-level reference. Check each tone at the planned ear position.

3. Record the timing and brain signals

- Choose the neural recorder and headstage, then match its event inputs to the tone-output system.
- Record tone markers in both sound and neural data. Check actual sound arrival and clock drift; a shared marker alone does not make two clocks identical.
- Include cables, power, fixed mounts, and data storage. Add a camera only if behavior is needed; align its frame times too.
- A soundproof box is not required by current team direction. Record background sound and the setup geometry.

Equipment candidates are from the forwarded BCM list; linked specifications were checked October 8, 2026. Mouse state, frequency range, and final hardware remain open.

Prices: sound delivery

Checked October 8, 2026; one new item/package per row

Prices are USD unless marked EUR. **Quote** means no public numerical price was verified. Shipping and taxes are extra unless stated by the seller. Choose one sound path; the rows are alternatives.

Item	Price	What it covers / status
TDT ES1	Quote	Speaker through the air; intended use from 4 kHz.
TDT EC1	Quote	Tubing/coupler speaker; alternative to ES1.
TDT ED1	Quote	Driver for ES1/EC1; needs waveform input.
TDT ZB1PS	Quote	Powered chassis for ED1 or SA1.
TDT MF1	Quote	SONIC speaker; specify package and check availability.
TDT SA1	Quote	SONIC amplifier; also needs zBus power.
Peerless XT25TG30-04	\$28.47	Bare tweeter; DigiKey showed zero stock.
Scan-Speak R2004/602200	\$178.60	Bare tweeter; stock unconfirmed.
ART SLA-1	\$279.99	Conventional amplifier candidate; in stock displayed.
NI USB-6361, 781442-01	\$3,030.42	Timed output/DAQ; dealer price includes duties/tariffs.
Digilent Analog Discovery 3	\$379.00	Optional bench waveform device; final playback suitability unverified.

What else goes with these?

- ES1/EC1 use ED1; MF1 uses SA1. Either TDT route needs zBus power. Confirm supplied speaker cables and stands before adding extra charges.
- Either bare tweeter needs a suitable conventional amplifier, wiring, and mounting. ART is a candidate; it does not drive ES1/EC1.
- NI and AD3 are output alternatives. AD3 sequence length and tone-marker timing need bench checks before use in the final setup.

Item names link to listings/docs. TDT pricing uses its quote route. Prices do not establish calibrated frequency coverage or equipment access.

Prices: sound measurement

Choose one complete microphone and recording path

Avisoft values are its published USD prices. **Edu** is a conditional educational price; Rice eligibility needs confirming.

Item / SKU	Price	Included / needed
116Hm base, 41165/41166	\$7,200 \$6,360 edu	Recorder interface; microphone extra; software included.
116Hm kit, 51165/51166	\$8,160 \$7,320 edu	Microphone/preamp, cable, stand, and software included.
CM16/CPMA, 40011	\$1,080	Mic kit for USG; already in the complete 116Hm kit.
Mic power module, 40017	\$1,320	Possible B&K-preamp-to-USG conditioning.
CM16/CPMA-5V, 40013	\$1,800	External-DAQ mic package; no anti-alias filter.
CM16/CPMA40-5V, 40014	\$2,160	Adds gain; no anti-alias filter.
CM16/CPMA48AAF-5V, 40026	\$2,580	Gain/filter package for compatible external acquisition.
40 kHz reference, 60105	\$552	One-frequency acoustic reference.
B&K 4939	Quote	Cartridge; preamp, conditioning, and ADC extra.
B&K 4939-A-011	Quote	Includes 4939 and 2670 preamplifier.
B&K 2670	Quote	Standalone preamp; omit if already in the set.
NEXUS 2690-A-0F2	Quote	Conditioner option; still needs acoustic ADC.
B&K 4231 calibrator	EUR 1,764	New listing; excludes VAT; availability to confirm.
DP-0775 adaptor	Quote	Quarter-inch adaptor for the 4939.

- Complete packages replace their included parts. Do not add a second microphone, preamp, or software license.
- The filtered 40026 + NI path is an alternative to a USG recorder. Its gain, wiring, sampling, and software still need checking.
- A 40 kHz or 1 kHz reference checks one frequency. We still need calibration information across our chosen tone range.

Prices: recording and data

Examples for functions we still need to provide

The Intan items show what a separate neural system could cost. Our implant and recorder choice remain open.

Item	Price	What remains
Intan C3004 RHD controller	\$9,950 list	Includes USB/power cables; supplier confirmation needed.
Intan C3314, 32-channel headstage	\$940 list	Check electrode/implant connector.
Intan C3203 SPI cable, 0.9 m	\$215 list	One headstage connection.
Electrode/probe and adapter	Model open	Choose with the team; not in Intan subtotal.
Dell Slim ECS1250, offer ecs1250_so_16	\$549.99 offer	Windows 11, i3, 8 GB/512 GB; monitor extra.
Samsung T7 Shield 2 TB, MU-PE2T0S/AM	\$574.99 listed	Delivery unavailable on Samsung page; reference only.
Monitor / PC upgrades / backup storage	Model open	Match acquisition load and data volume.
Basler acA1300-200um camera	\$585.00	Optional body; lens, USB/I/O cables, light, and mount extra.
Camera accessories	Model open	Price after field of view and geometry are chosen.
Intan RHX software	Free	Hardware is priced separately.
Digilent WaveForms	Free	For the AD3 bench path.
RECORDER USGH	Included	Already counted with USG hardware.
Stimulus/control and analysis software	Tool open	Development time and any paid license extra.

- These are comparison prices, not selected models. Match the computer and headstage to the actual recording plan.
- Record tone markers in the neural data. Use suitable acoustic acquisition for microphone waveforms; neural inputs do not establish ultrasound capture.
- The sound, acoustic, neural, and optional video clocks still need an alignment check.

Smaller parts and subtotals

Final quantities depend on our connection diagram

Part / support	Price	How to count it
Pomona 2249-C-36 BNC cable	\$30.99 each	Example 0.9 m cable; count from wiring map.
Avisoft 52002 DIN adapter	\$48 each	USG timing connection option.
Avisoft 50021 XLR-5 cable, 2 m	\$60 each	Extra only if kit cable is unsuitable.
On-Stage DS7200B stand	\$15.95 each	Extra only if needed; clamp/adapter separate.
TDT speaker cable	Check package	May already be supplied.
Speaker wiring / output-device adapters	Model open	Match chosen amplifier and DAC connectors.
Clock/level interface, if needed	Model open	Match recorder inputs before pricing.
Fixed mounts / clamps / base / baffle	Design open	Choose after geometry; omit bundled stands.
Animal support equipment	Preparation open	Depends on later awake/sedated decision.
Enclosure / acoustic treatment	Optional	Soundproof box not required by current direction.

Partial subtotals of priced parts

One of each named item	Subtotal
NI + 40026 filtered microphone package + 40 kHz reference	\$6,162.42
NI + complete 116Hm kit + 40 kHz reference	\$11,742.42
Same, using an eligible educational 116Hm kit	\$10,902.42
Intan controller + 32-channel headstage + SPI cable	\$11,105 list
Peerless + ART amplifier	\$308.46
Scan-Speak + ART amplifier	\$458.59

- These sums cover only the named parts. Add remaining speaker, neural, computer, wiring, mounting, and calibration costs for the chosen path.
- Quote-only and model-open items are unpriced; they are not zero-cost items. Tax, shipping, and integration work are additional.

Public prices checked October 8, 2026. No supplier quote, equipment order, or measured compatibility is implied.